STRATEGIC CAPITAL PORTAL

FARMSENSE SEED ROUND A-1

Executive Summary

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FarmSense Investor Packet
Seed Round A-1

COVER PAGE

FarmSense™ Global Water Security Infrastructure

Seed Round A-1: $25M at $75M Post-Money

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1. EXECUTIVE SUMMARY

FarmSense is the deterministic operating system for global water security.

The Crisis: Agriculture consumes 70% of freshwater. 40 countries face acute water scarcity by 2030. Water courts require evidence that doesn't exist.

The Catalyst: Colorado's San Luis Valley — $500/acre-foot groundwater fees, 89,000 AF annual depletion, 300,000 acres at risk.

The Solution: A sensor-to-satellite mesh network with cryptographic chain of custody. Legally admissible in Water Court. 20-30% water reduction. 18-22% ROI increase.

The Model: Government grants fund 100% of infrastructure. Farmers get free compliance. Premium tiers (20m, 10m, 1m resolution) generate 100% margin SaaS revenue.

The Ask: $25M Seed Round at $75M post-money.

The Outcome: 5% global penetration (400,000 fields) by 2036 = $235M ARR at 100% margin. Exit valuation: $35-70B. Founder's 35% = $12-25B.

2. THE OPPORTUNITY

The $500/AF Catalyst

The Rio Grande Basin faces collapse:
Annual aquifer depletion: 89,000 acre-feet
Groundwater fees: $500/acre-foot (quadrupled from $75-150/AF)
Reservoir storage: 26% of historical capacity
Irrigated agriculture: 300,000 acres at risk

Economic Impact per 126-acre field:

20% water reduction = 50.4 AF saved
50.4 AF × $500/AF = $25,200 annual savings
FarmSense Enterprise: $499/month = $5,988/year
First-year ROI: 421%

Global Context:

634M acres irrigated cropland worldwide
40 countries facing acute water scarcity by 2030
Interstate litigation threatening federal takeovers
No existing solution provides legally defensible water accounting
FarmSense creates the infrastructure layer that makes water management enforceable, measurable, and optimizable.

3. THE SOLUTION

FS-1: Deterministic Farming Operating System

FarmSense replaces stochastic "intuition" with computational precision:

Dimension	Current Standard	FarmSense DFOS
Observation	Visual scouting / manual probe	1m centimetric Kriging + ground-truth
Calculation	Crop-coefficient (static)	SPAC thermodynamics (real-time)
Decision	Intuition-based schedules	Edge-calculated VRI prescription
Evidence	Paper logs, human memory	SHA-256 sovereign ledger
Resolution	Field-scale variable	1m grid, legally admissible

Hardware Ecosystem:

Tier	Device	Function
Regional	RSS (40ft container)	Territory master, blackout-proof
District	DHU (35ft pole)	100-pivot mesh manager
Field	PMT (pivot-mounted)	Edge aggregator, 50m Kriging
Sensors	VFA/LRZ/PFA	Ground-truth collection

Key Metrics:

Water reduction: 20-30%
ROI increase: 18-22%
Grid resolution: 50m → 1m (400x improvement)
Legal admissibility: Water Court ready

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4. MARKET SIZE

TAM/SAM/SOM Analysis

Market Tier	Size	Definition
Total Market	$3-5T	Global agriculture + water infrastructure
Precision Ag	$12.8B	Irrigation tech market by 2030
FarmSense TAM	$19-44B	Infrastructure + financial derivatives

Global Penetration Target:

Region	Fields	Timeline
SLV Subdistrict 1	1,280	2026-2027
Rio Grande Basin	6,000	2027-2028
Colorado River Basin	25,000	2028-2029
High Plains Aquifer	50,000	2029-2030
Top 10 Watersheds	200,000	2030-2034
5% Global	**400,000**	**2036**

Revenue Layers (Annual TAM):

Layer	Value	Margin
Core SaaS	$2.94B	100%
Water derivatives	$5-10B	75%
Carbon credits	$2-5B	90%
Insurance data	$3-8B	85%
Commodity futures	$5-15B	80%
Regulatory licensing	$1-3B	90%
Total TAM	**$19-44B**	**82% blended**

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5. BUSINESS MODEL

G2C Infrastructure: Government-to-Consumer

Farmers never pay for compliance. They pay only for optimization.

Tier	Resolution	Price	Who Pays
Base	50m	**FREE**	Government grants
Plus	20m	$99.99/mo	Farmers (optional)
Ultra	10m	$299.99/mo	Farmers (optional)
Enterprise	1m + 10cm	$499.99/mo	Large operations

Unit Economics (100% Margin):

Metric	Value
Blended ARPU	$588/field/year
Year 5 fields	52,000

Metric	Value
Year 5 ARR	$30.6M
Gross margin	100%
LTV:CAC	Infinite

Infrastructure Arbitrage: Grants deploy the network. Paid tiers capture pure margin.

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6. FINANCIAL PROJECTIONS

The $10-100B Path

Core SaaS Revenue:

Year	Fields	Paid %	ARR	Margin
1	320	15%	$28K	100%
2	2,560	12%	$180K	100%
3	10,240	10%	$602K	100%
5	52,000	10%	$3.06M	100%
10	400,000	10%	$235M	100%

Ancillary Revenue (Year 10):

Stream	Annual Value
Core SaaS	$235M
Carbon credits	$50-100M
Insurance data	$100-200M
Commodity futures	$200-500M
Water derivatives	$500M-1B
Regulatory licensing	$50-200M
Total	**$1.14-2.34B**

Valuation Scenarios (Year 10):

Scenario	Multiple	Valuation	Founder 35%
Base	50x	$11.7-23.4B	$4.1-8.2B
Optimistic	100x	$23.4-46.8B	$8.2-16.4B
Sovereign	150x	$35.1-70.2B	$12.3-24.6B

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7. DEVELOPMENT TIMELINE

Validation-Driven Parallel Expansion

Phase	Year	Scope	Milestone
Proof	2026	CSU SLV 2-Field Pilot	20%+ savings, legal admissibility
Capture	2026-27	SLV Subdistrict 1 (1,280)	Water Court precedent
Scale	2027-28	Rio Grande Basin (6,000)	3-state expansion

Phase	Year	Scope	Milestone
National	2028-29	Colorado River (25,000)	Interstate compact
Global	2030-34	Top 10 Watersheds (200,000)	World Bank credibility
Dominance	2034-36	5% Global (400,000)	Network effects mature

Grant Pipeline:

Program	Amount	Deadline
USDA SBIR Phase I	$300K	Mar 2026
USDA SBIR Phase II	$1.1M	Sep 2026
NRCS CIG	$75K-$5M	Apr 2026
DOE Water-Energy	$2M-$10M	Jun 2026
USAID Global Dev	$50M	Jul 2026
Gates Foundation	$100M	Aug 2026
World Bank IDA	$75M	Sep 2026

Total Pipeline: 160+ grants, **$810M+** over 2-3 years

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8. COMPETITIVE MOAT

Determinism vs. Stochastic Estimation

Competitor	Approach	Weakness	Our Advantage
CropX	Cloud-only	Latency, connectivity	Edge + sovereign ledger
FieldNET	Pivot control only	No soil/VRI	Full SPAC + 1m VRI
Arable	Weather-only	No ground-truth	Sensor + satellite fusion
Sat-only	Trend analysis	No deep-profile	1m Kriging + 48" VFA

Moat Durability:

Data Moat: 4+ years ground-truth calibration per field

Regulatory Moat: Legal recognition as “approved monitoring method”

Network Moat: 400,000-field mesh effects

Infrastructure Moat: Government-funded deployment = switching costs

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9. USE OF PROCEEDS

$25M Seed Round Allocation

Category	Allocation	Amount
Legal entity setup	2%	$500K
Operations reserve	8%	$2M
Grant application engine	5%	$1.25M
Hardware prototype	15%	$3.75M
Field operations team	20%	$5M

Category	Allocation	Amount
Software/Kriging engine	15%	$3.75M
Regulatory/Water Court	10%	$2.5M
Business development	10%	$2.5M
Reserve	15%	$3.75M

The Seed Unlocks:

Immediate operations (no 9-12 month grant delay)
AI-accelerated 160+ grant applications in 6 weeks
Multi-state and international pilots
Water Court evidence preparation

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10. RISK FACTORS

Technical:

Sensor failure (mitigated: IP67, redundant design)
RF interference (mitigated: CSS LoRa, mesh topology)
Kriging accuracy (mitigated: 4+ year calibration, MAPE <5%)

Market:

Farmer adoption (mitigated: free compliance tier)
Competitive response (mitigated: regulatory moat)
Regulatory rejection (mitigated: CSU validation)

Execution:

Grant timing (mitigated: seed bridge, 18-month runway)
Team scaling (mitigated: clear roadmap, equity incentives)
International complexity (mitigated: phased rollout)

Confidence Level: 85-90% achievable with seed + AI-accelerated grants

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11. THE EXIT

Exit Timeline: 2036 (Year 10)

Strategic Buyers:

Buyer	Rationale	Multiple
Trimble	Precision ag portfolio gap	50-75x
Deere	Farm automation ecosystem	60-80x
Bayer	Crop science integration	50-70x
Microsoft/Google	Climate data infrastructure	75-100x
Sovereign Wealth	Water security strategic asset	80-120x

Exit Math (Conservative):

Metric	Value
Year 10 ARR	$235M core + $1B ancillary = **$1.24B**
Infrastructure multiple	**50-100x**
Exit valuation	**$62-124B**
Founder stake (35%)	**$21.7-43.4B**

Metric	Value
Angel stake (0.35%)	**$217-434M (8,680-17,360x)**

This is Visa/Mastercard for water. The financial infrastructure layer for the world's most scarce resource.

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12. GLOSSARY & INDEX

A

Adaptive Recalculation Engine — Dynamic computation scheduler. Modes: DORMANT, ANTICIPATORY, FOCUS RIPPLE, FOCUS COLLAPSE.

AES-256 — Advanced Encryption Standard. Used for all field telemetry.

AllianceChain — FarmSense's PBFT consensus protocol for Digital Water Ledger.

Alpha-Sled — Removable electronics cartridge deployed seasonally.

ANTICIPATORY Mode — Engine state triggered by rapid temperature rise.

Aquifer — Underground geological formation containing groundwater.

B

Black Box Cache — 128GB SSD in DHU for 30-day encrypted audit recording.

BLE 5.0 — Bluetooth Low Energy for technician maintenance access.

BOM — Bill of Materials. Complete parts list with costs.

C

CSS LoRa — Chirp Spread Spectrum LoRa at 915MHz. Primary telemetry protocol.

CSA — Corner-Swing Arm. BLE 5.2 distance ranging device.

COLLAPSE Mode — Critical engine state during active irrigation.

Cokriging — Multivariate geostatistical interpolation.

Core Compute Server (Zo) — Cloud-based scientific engine.

Crop Coefficient (Kc) — Factor relating actual ET to reference ET.

D

DHU — District Hub. Level 2 infrastructure node. 35' pole, Orin Nano, 100-pivot radius.

Digital Water Ledger (DWL) — Cryptographically signed chain of custody.

DIL — Data Integration Layer.

DORMANT Mode — Low-power engine state. 4-hour polling, 8µA draw.

E

Edge-EBK — Empirical Bayesian Kriging at PMT. 50m resolution.

ESP32-S3 — MCU in PMT. Dual-core, vector FPU, LoRa.

ET (Evapotranspiration) — Combined water loss from soil + plants.

F

FOCUS RIPPLE — Engine state triggered by >5% moisture deviation.

G

GNSS — Global Navigation Satellite System. Sub-cm positioning.

GroPoint Profile — Commercial soil moisture sensor in VFA.

H

HDPE SDR9 — High-Density Polyethylene pipe for VFA/LRZ shells.

Hydraulic Conductivity (Ksat) — Rate of water movement through saturated soil.

I

IMU — Inertial Measurement Unit. 9-axis for tower stability.

ISLAND Mode — PMT autonomous operation when DHU offline.

K

Kc — See: Crop Coefficient.

Kriging — Geostatistical interpolation. FarmSense uses Ordinary, Regression, Edge-EBK.

L

LiFePO4 — Lithium Iron Phosphate battery. DHU, PFA.

LiSOCl2 — Lithium Thionyl Chloride. LRZ1/LRZ2 (10-year life).

LoRa — Long Range low-power wireless. 915MHz CSS LoRa.

LRZ1/LRZ2 — Lateral Root-Zone Basic/Reference. $29/$54 spatial scouts.

LTE-M — Low-power cellular for DHU backhaul fallback.

M

MAD — Management Allowable Depletion. % of available water before crop damage.

MAPE — Mean Absolute Percentage Error. Kriging validation target <5%.

N

NREP — Non-Repudiable Evidence Prime. Legal admissibility standard.

P

PBFT — Practical Byzantine Fault Tolerance. Consensus algorithm.

PMT — Pivot Motion Tracker. Field hub with edge-EBK engine.

PFA — Pressure & Flow Anchor. Flow measurement node.

PSLC — Pseudo-SLC flash. High-endurance SSD in DHU.

R

RFM95W — 915MHz LoRa radio module.

RSS — Regional Superstation. 40ft container, Threadripper, 50TB.

RTK — Real-Time Kinematic. Sub-cm GNSS positioning.

S

SDR9 — Standard Dimension Ratio 9. HDPE pipe specification.

SLV — San Luis Valley. Colorado pilot location.

SPAC — Soil-Plant-Atmosphere Continuum. Thermodynamic model.

T

TAM — Total Addressable Market. FarmSense TAM: $19-44B/year.
TimescaleDB — PostgreSQL extension for time-series data.
V
VFA — Vertical Field Anchor. 48" deep-profile truth node.
VRI — Variable Rate Irrigation. Prescription-based water application.

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